

PROJECT (Group B): Sales & Inventory Management System

PART 1: Database Creation Questions

- Create a database named sales_inventory_system.
- Create a customers' table with customer_id, first_name, last_name, email, phone, city.
- Create a suppliers table with supplier_id, supplier_name, contact_email.
- Create a products table with product_id, product_name, category, price, supplier_id.
- Create an orders table with order_id, customer_id, order_date.
- Create an order_details table with order_detail_id, order_id, product_id, quantity.
- Create a payments table with payment_id, order_id, payment_date, amount, payment_method.

PART 2: Data Insertion Questions

- Insert at least 20 customers.
- Insert at least 10 suppliers.
- Insert at least 30 products.
- Insert at least 40 orders.
- Insert at least 60 order_details.
- Insert at least 40 payments.
- Ensure each order has at least one product.
- Ensure each product belongs to one supplier.
- Ensure each order has one payment record.

PART 3: SQL Query Questions

- Display all customers and their orders.
- List all products with their suppliers.
- Show total number of orders made by each customer.
- Find the total sales amount per day.
- Find the total revenue generated.
- List the top 5 best-selling products.
- Show customers who spent more than a specified amount.
- Find the average order value.
- Display products that have never been sold.
- Find customers who have never placed an order.
- Show orders that contain more than 3 products.

PART 4: Intermediate SQL Questions

- Rank customers based on total amount spent.
- Find the product category with the highest revenue.
- Find the supplier whose products sell the most.
- Display customers who placed more than 5 orders.
- Find orders with the highest total quantity of items.
- Show monthly sales performance.
- Find products with price above the average product price.
- Find customers whose total spending is above the overall average spending.

PART 5: Business & Reporting Questions

- Generate a report showing customer name, order date, product name, quantity, and amount paid.
- Generate a summary report of total customers, total products, total orders, and total revenue.
- Identify loyal customers (customers with highest number of orders).
- Identify slow-moving products (products with very low sales).
- Recommend products that should be restocked based on sales.
- Recommend the best performing supplier.

FINAL DELIVERABLE

- Submit SQL file for table creation.
- Submit SQL file for data insertion.
- Submit SQL file containing answers to all questions.
- Submit a short presentation explaining database design, key queries, and business insights.